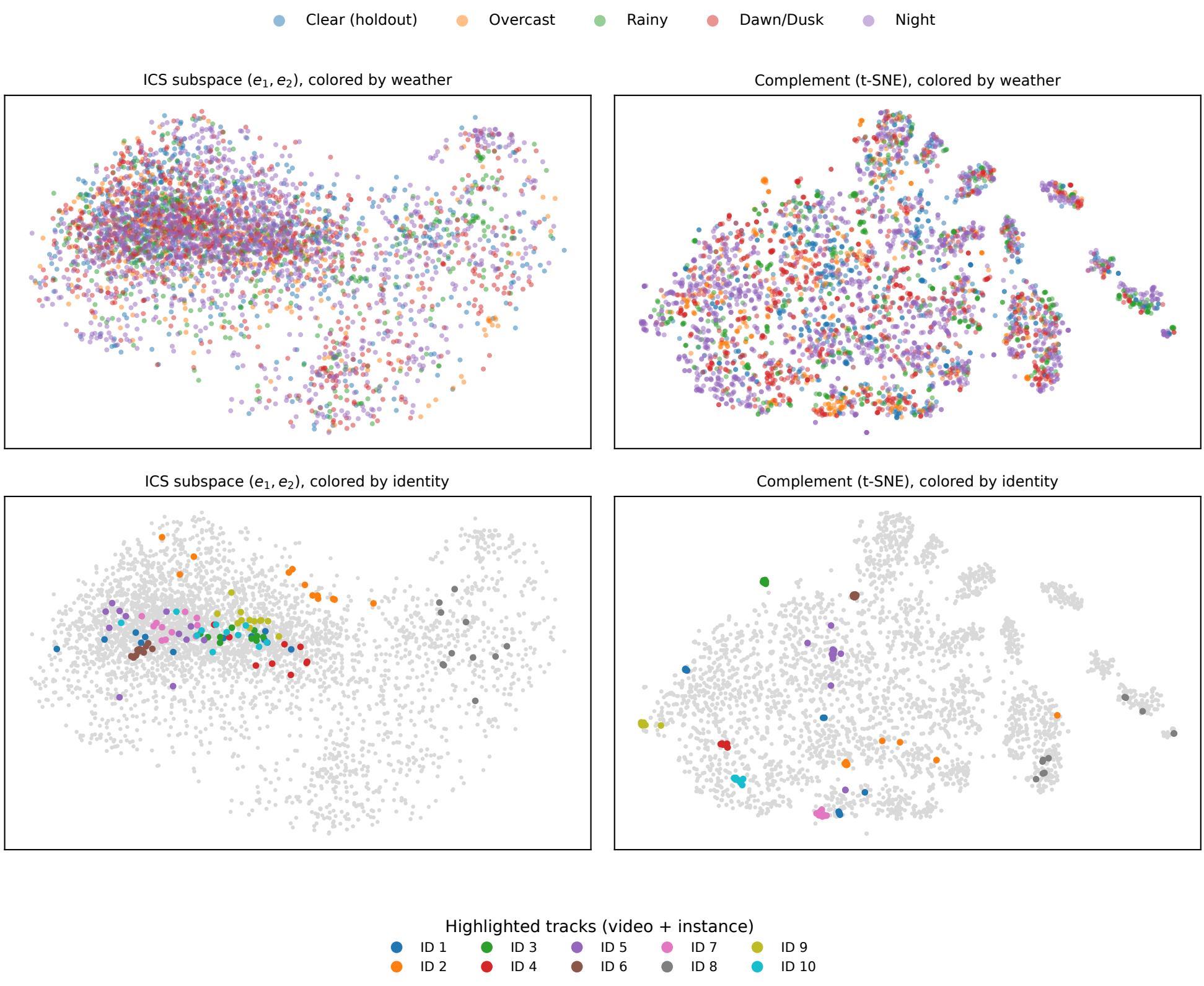
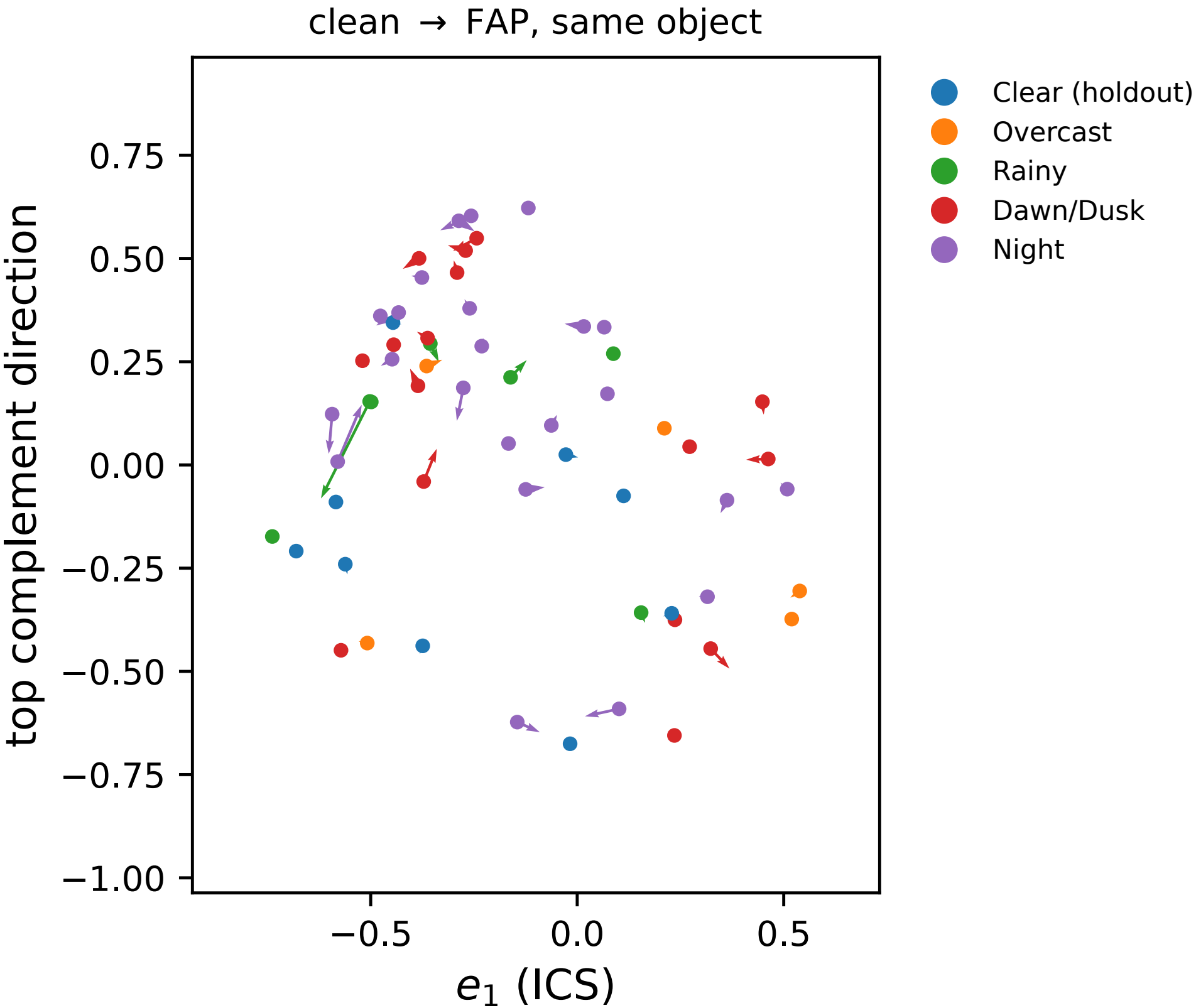
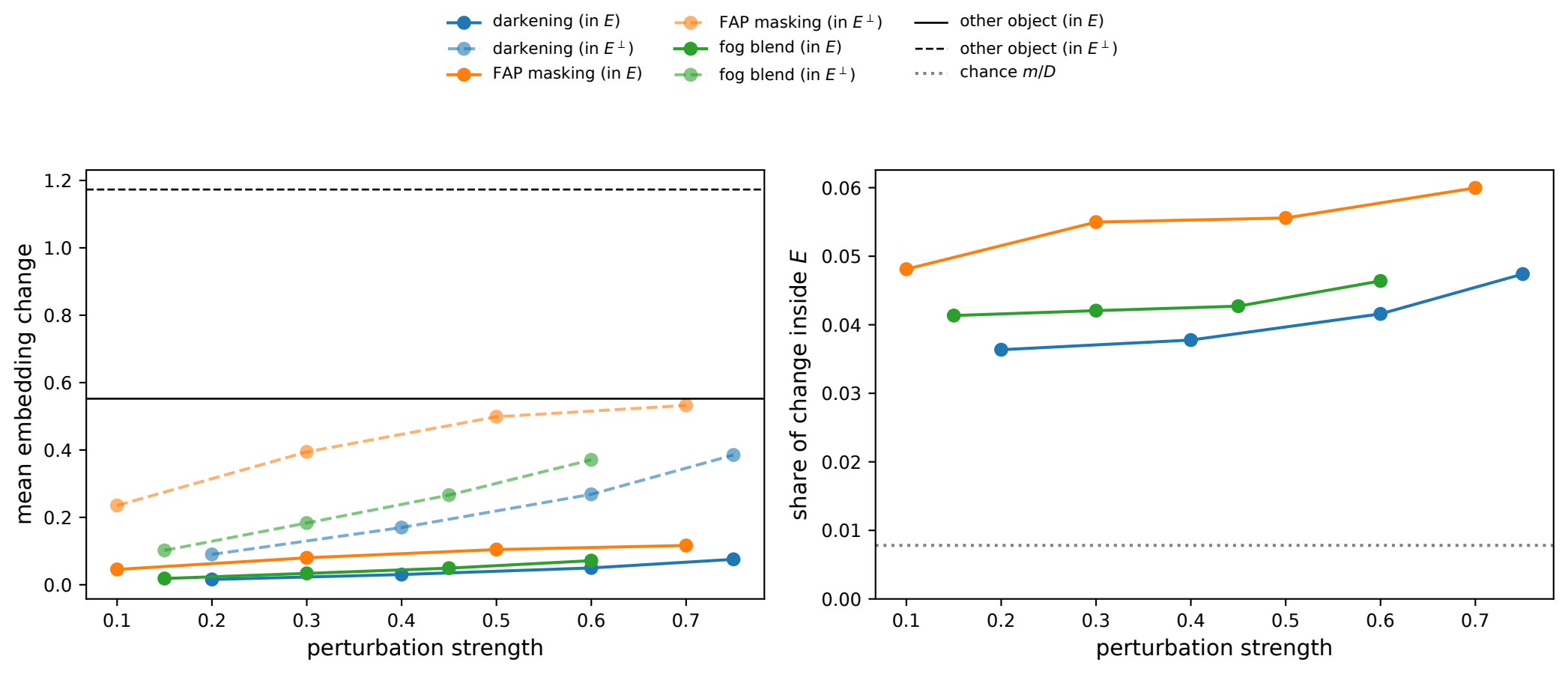
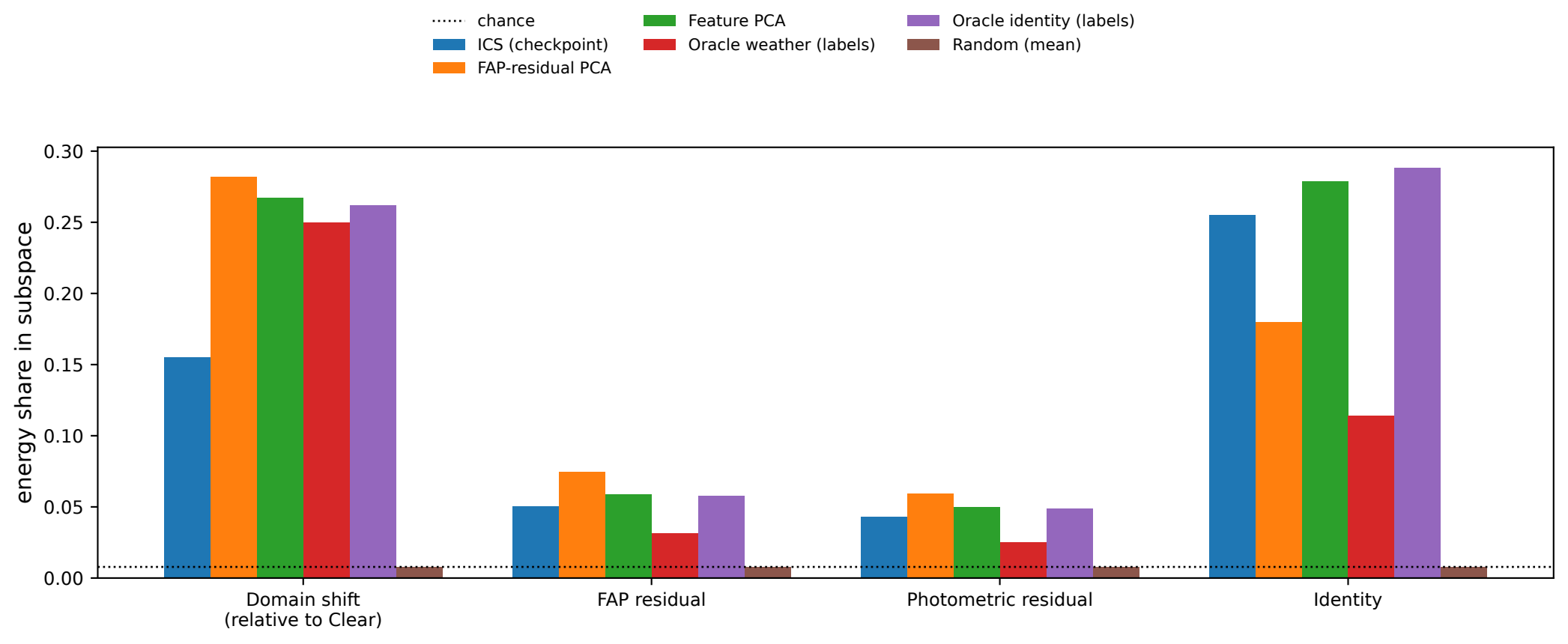

Clear (holdout)
Overcast
Rainy
Dawn/Dusk
Night
ICS subspace (e_1, e_2), colored by weather
Complement (t-SNE), colored by weather
ICS subspace (e_1, e_2), colored by identity
Complement (t-SNE), colored by identity
Highlighted tracks (video + instance)
ID 1
ID 2
ID 3
ID 4
ID 5
ID 6
ID 7
ID 8
ID 9
ID 10

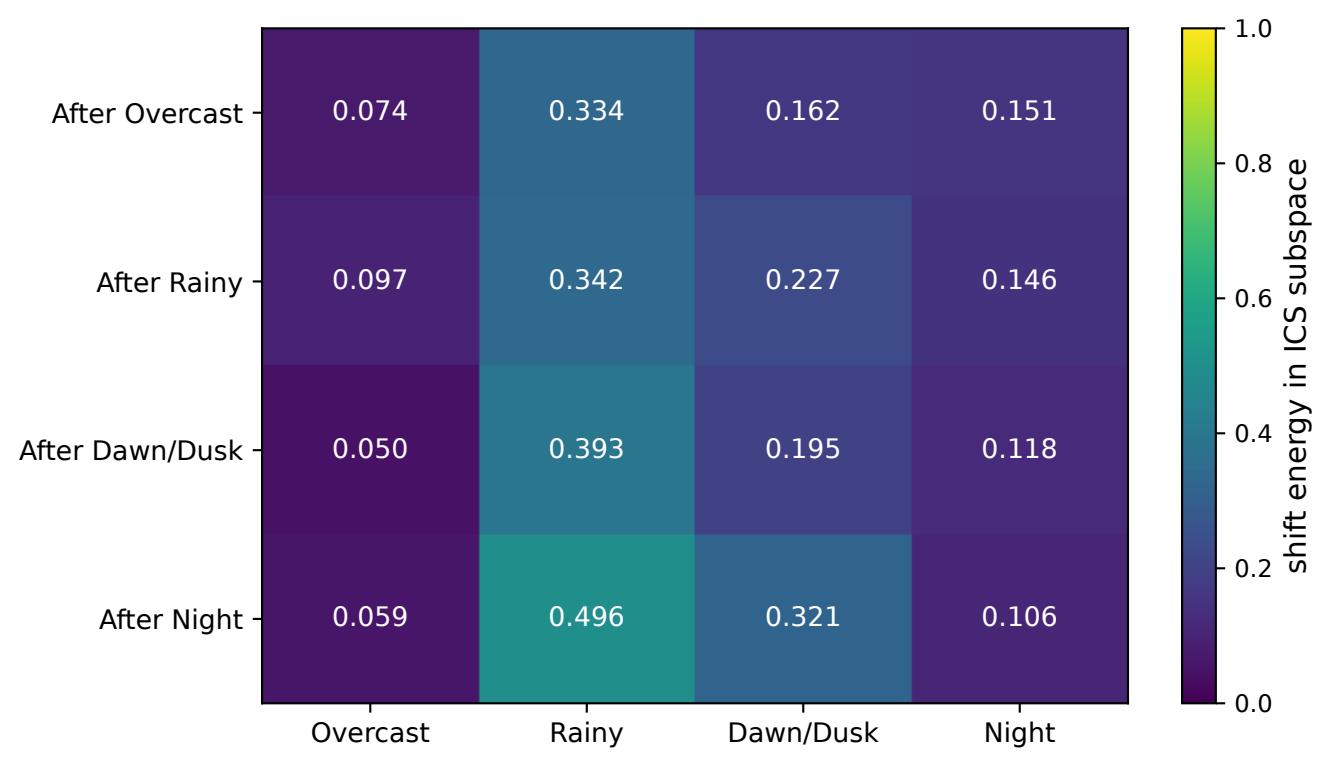

clean → FAP, same object
Clear (holdout)
Overcast
Rainy
Dawn/Dusk
Night
top complement direction
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00
−0.5
0.0
0.5
e_1 (ICS)


low e_1
Overcast
-0.803
Night
-0.802
Night
-0.799
Night
-0.799
Clear
-0.797
Dawn/Dusk
-0.793
Overcast
-0.793
Night
-0.793
Night
-0.793
Night
-0.792
high e_1
Dawn/Dusk
+0.750
Clear
+0.733
Dawn/Dusk
+0.723
Overcast
+0.722
Night
+0.718
Dawn/Dusk
+0.716
Clear
+0.715
Night
+0.711
Clear
+0.711
Clear
+0.708


low e_2
Dawn/Dusk
-0.889
Dawn/Dusk
-0.850
Night
-0.847
Overcast
-0.841
Clear
-0.837
Clear
-0.830
Night
-0.830
Rainy
-0.829
Rainy
-0.827
Clear
-0.824
high e_2
Night
+0.767
Night
+0.751
Dawn/Dusk
+0.735
Dawn/Dusk
+0.732
Dawn/Dusk
+0.730
Dawn/Dusk
+0.728
Dawn/Dusk
+0.722
Overcast
+0.722
Dawn/Dusk
+0.716
Night
+0.715


darkening (in E)
darkening (in $E^\perp$)
FAP masking (in E)
FAP masking (in $E^\perp$)
fog blend (in E)
fog blend (in $E^\perp$)
other object (in E)
other object (in $E^\perp$)
chance m/D
mean embedding change
perturbation strength
share of change inside E
perturbation strength


chance
ICS (checkpoint)
FAP-residual PCA
Feature PCA
Oracle weather (labels)
Oracle identity (labels)
Random (mean)
energy share in subspace
0.30
0.25
0.20
0.15
0.10
0.05
0.00
Domain shift
(relative to Clear)
FAP residual
Photometric residual
Identity


After Overcast
After Rainy
After Dawn/Dusk
After Night
0.074
0.334
0.162
0.151
0.097
0.342
0.227
0.146
0.050
0.393
0.195
0.118
0.059
0.496
0.321
0.106
Overcast
Rainy
Dawn/Dusk
Night
shift energy in ICS subspace
1.0
0.8
0.6
0.4
0.2
0.0